Exact Graph Structure Learning with Tensor Affinity for Topology-Refined Deep Clustering

YU WANG*, South China University of Technology, China
JUNYU LI*, South China University of Technology, China
HONGMIN CAI†, South China University of Technology, China
ZHUOYAO WANG, Pengcheng Laboratory, China
YIQUN ZHANG†, South China University of Technology, China

Materializing clean semantic communities from graph databases motivates Deep Graph Clustering (DGC) to disentangle node embeddings, driving the exact refinement of corrupted topologies. Existing DGC methods relying on static structures struggle to adaptively filter the spurious linkages within these corrupted topologies, whereas fixed high-order methods incur prohibitive computational bottlenecks when capturing complex semantics. To address these limitations, we propose **TAGSL** as a unified DGC framework integrating **T**opology-aware tensor **A**ffinity with adaptive **G**raph **S**tructure **L**earning to materialize clean semantic communities. Our tensor affinity captures high-order interactions by modulating interaction strengths with local structural context to reduce the influence of spurious linkages. Concurrently, the learned structural operator adapts to evolving embedding geometry under spectral rank constraints to achieve exact graph structural decoupling and support semantic community formation. Our optimization scheme uses a separable Nyström approximation to avoid dense tensor construction and Ky Fan reformulation to make the exact rank constraint tractable, enabling practical deployment in graph database systems. Extensive experiments on seven diverse graphs against thirteen baselines demonstrate strong clustering performance, with embedding analyses showing more separable semantic communities. Runtime and memory analyses further demonstrate computational efficiency and approximately linear scaling trends at the million-node scale. Moreover, a real-world multi-omics case study shows reduced pathway mixing and relatively distinct semantic views that are qualitatively consistent with domain expert knowledge. Our code is publicly available at https://github.com/wy-1997/TAGSL.

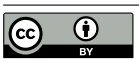
CCS Concepts: • **Information systems → Clustering**.

Additional Key Words and Phrases: Deep Graph Clustering, High-Order Representation, Adaptive Structure Learning, Topology Refinement, Scalable Clustering

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*These authors contributed equally to this work.
†Hongmin Cai and Yiqun Zhang are corresponding authors.

Authors' Contact Information: Yu Wang, South China University of Technology, Guangzhou, China, yuuwang97@gmail.com; Junyu Li, South China University of Technology, Guangzhou, China, chunyulee@scut.edu.cn; Hongmin Cai, South China University of Technology, Guangzhou, China, hmcai@scut.edu.cn; Zhuoyao Wang, Pengcheng Laboratory, Shenzhen, China, wangzhy01@pcl.ac.cn; Yiqun Zhang, South China University of Technology, Guangzhou, China, yqzhangzyq@gmail.com.

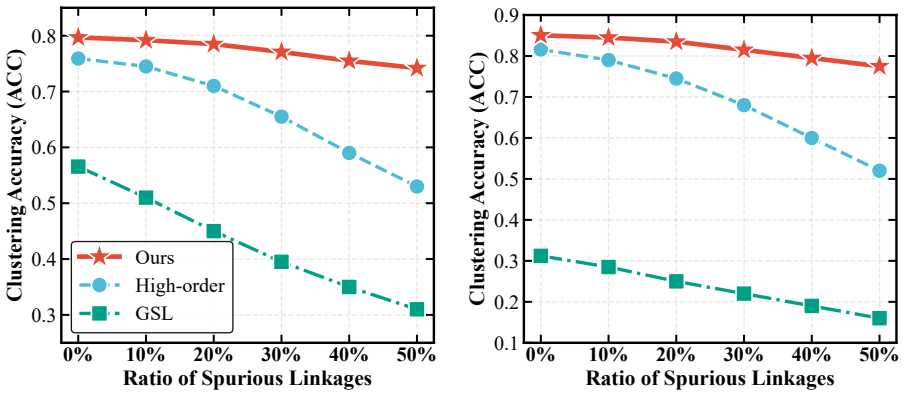


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1 Introduction

Discovering communities in graph data is an important graph analytics task in data management [12, 13]. Classical spectral clustering identifies communities through eigendecomposition of affinity matrices or graph Laplacians [29, 34]. To capture semantic structure through learned representations, Deep Graph Clustering (DGC) learns node representations and partitions graph nodes into semantically coherent communities in an unsupervised manner [40, 43]. Many DGC methods employ Graph Neural Networks (GNNs) to encode graph topology and node attributes into embeddings that reflect latent community structure [27, 41, 56].

Existing DGC methods adopt different strategies for handling graph topology. *Reconstructive methods* [20, 41] use autoencoders to reconstruct node features or the observed topology. Constructing graph databases from multi-source records or relational tables can introduce spurious linkages into the observed topology [22, 39]. Reconstructive methods risk retaining spurious linkages during representation learning. As an alternative to topology reconstruction, *contrastive learning approaches* [35, 59, 60] learn discriminative representations by maximizing agreement across augmented graph views. Augmented graph views still depend on the fixed observed topology and remain sensitive to structural corruption. Seeking to alleviate dependence on a fixed observed topology, *graph structure learning* (GSL) methods can refine graph structure by regulating graph connectivity and modeling node relations [23, 30, 32]. Exact Laplacian rank constraints further regulate graph connectivity by enforcing a prescribed number of connected components [23, 30]. Optimization under exact Laplacian rank constraints typically relies on eigendecomposition, which increases computational cost on large graphs. Node relations can be modeled through pairwise affinities, leaving multi-node interactions implicit. Beyond pairwise relations, *high-order methods* [6, 10, 58] model multi-node interactions through structural motifs or tensor affinities. Extending relation modeling from pairwise to multi-node interactions can increase construction and storage costs of high-order structures [6, 10].



(a) ACC vs. Spurious Ratio @ Cora (b) ACC vs. Spurious Ratio @ Flickr

Fig. 1. Clustering performance under increasing ratios (0%–50%) of spurious linkages on Cora and Flickr. Representative GSL methods [16] and high-order models [49] show decreasing clustering accuracy with increasing spurious linkages. Our method shows a more gradual decline and higher ACC across the tested ratios on both graphs. The trend is consistent with jointly modeling high-order interactions and refining graph topology under structural corruption.

Structural corruption in real-world graph databases motivates both explicit control of graph connectivity and richer modeling of node interactions. Exact Laplacian rank constraints enforce the prescribed number of connected components. High-order modeling captures multi-node interactions beyond pairwise relations. However, integrating exact structural constraints and high-order interactions into adaptive topology learning presents scalability challenges for DGC. Spurious linkages perturb the observed graph connectivity and local neighborhood relations. The resulting corrupted neighborhoods alter the structural signals propagated during representation learning. Topology refinement consequently involves distinguishing reliable graph connectivity from spurious linkages. Fig. 1 examines whether learned communities remain robust as spurious linkages accumulate on the homophilic Cora and heterophilic Flickr graphs. The ratio of spurious linkages is varied from 0% to 50% by randomly adding links to the observed topology. As illustrated in Fig. 1, representative GSL and high-order baselines [16, 49] show an overall decline in clustering performance (ACC) as the ratio of spurious linkages increases. Sensitivity to increasing spurious linkages is observed in the homophilic and heterophilic settings. Robust topology refinement under spurious linkages calls for coupling high-order interaction modeling with exact graph structural decoupling. Meanwhile, efficient realization of joint refinement requires a computationally scalable optimization strategy for adaptive graph structure learning.

Therefore, we propose TAGSL as a unified DGC framework integrating **T**opology-aware tensor **A**ffinity with adaptive **G**raph **S**tructure **L**earning to achieve exact graph structural decoupling. Specifically, TAGSL first introduces a topology-aware tensor affinity to capture high-order multi-way interactions. The tensor affinity modulates interaction strengths according to local structural context to account for variations in graph densities. The context-aware modulation reduces the influence of spurious linkages during high-order interaction modeling, consistent with the robustness trend in **Fig. 1**. Concurrently, TAGSL learns a structural operator that adapts to the evolving embedding geometry under exact spectral constraints. The prescribed rank condition fixes the null-space dimension at the target community count. The resulting exact graph structural decoupling supports the formation of semantic communities. To overcome computational bottlenecks, TAGSL combines a separable Nyström approximation with Ky Fan reformulation. The Nyström approximation avoids explicit dense tensor construction in high-order modeling, while Ky Fan reformulation makes the exact rank constraint tractable. With a fixed number of landmarks, the core high-order tensor computation scales linearly with the node count. Comprehensive evaluations on diverse graph benchmarks against 13 representative baselines demonstrate strong clustering performance. Further embedding analyses on representative heterophilic and homophilic graphs show more clearly separated semantic communities. Runtime and memory analyses demonstrate computational efficiency, with empirical scaling trends consistent with the complexity analysis. Beyond standard benchmarks, a real-world multi-omics case study demonstrates the applicability of TAGSL. The multi-omics results show reduced pathway mixing and relatively distinct semantic views that are qualitatively consistent with domain expert knowledge. **The main contributions are summarized as follows**:

- **A topology-aware tensor affinity** captures high-order interactions under structural corruption. Local structural context modulates interaction strengths across varying graph densities and reduces the influence of **spurious linkages**.
- **An adaptive graph structure learning mechanism** is developed under exact spectral constraints. The rank condition fixes the null-space dimension to the target community count, supporting **exact graph structural decoupling**.

- **An alternating optimization scheme** integrates a separable Nyström approximation with Ky Fan reformulation to avoid dense tensor construction while making the exact rank constraint tractable, supporting **linear scalability**.
- **Extensive evaluations** on seven benchmarks against 13 baselines demonstrate strong clustering performance and computational efficiency. A real-world multi-omics case study further shows reduced pathway mixing and semantic views qualitatively consistent with domain knowledge.

2 Related Work

Existing DGC methods explore diverse paradigms to extract semantic communities. **Reconstructive methods** prioritize local structural fidelity by minimizing topological and feature reconstruction errors. GAE and VGAE [20] reconstruct graph connectivity through inner-product decoding, whereas MGAE [41] marginalizes corrupted node features to learn graph representations. Subsequent approaches [8, 53, 56] introduce probabilistic clustering priors, adaptive graph convolutions, and sample-stability constraints, while multi-view extensions [31, 37] integrate view-specific representations and graph structures through adaptive fusion. Reconstructive frameworks preserve observed structures and attributes, with discriminative community boundaries typically encouraged by additional clustering or structural mechanisms. Seeking to enhance representation discriminability, **contrastive learning** shifts the focus toward representation consistency. Core frameworks including GRACE [59], GCA [60], MVGRL [15], and HeCo [44] learn representations by maximizing agreement across complementary graph views. Recent graph and deep clustering studies [25, 36, 52] explore topology-aware modeling and informative sample mining. The quality of contrastive representations depends on graph views that preserve informative topology and node semantics. Explicit structural objectives complement cross-view agreement by directly guiding topology refinement toward cluster-consistent graph structures.

Graph structure learning reduces reliance on **predefined static graphs** by dynamically refining network topology. Neural DGC methods jointly optimize structural representations and clustering objectives [2, 19, 32]. A distinct line of graph learning imposes explicit spectral constraints on graph connectivity. CLR [30] constrains the graph Laplacian rank to obtain exactly the specified number of connected components. Enforcing Laplacian rank constraints through eigendecomposition can be computationally expensive on large graphs. Recent methods [23, 42] reduce dependence on explicit spectral decomposition through orthogonality-constrained representations or structured doubly stochastic graphs. Affinity-based graph structure learning primarily models pairwise relations, leaving higher-order **multi-node interactions** implicit.

High-order methods explicitly model **multi-node interactions** beyond pairwise affinities. One line incorporates motifs and other higher-order structural patterns into graph representation learning [10, 55, 58]. Another line encodes multi-way dependencies through hypergraphs [13, 14, 47] or tensor-based formulations [6, 48]. Explicitly modeling multi-way relations can introduce substantial computational overhead on large graphs. High-order structures can be constructed before representation learning or refined jointly during training [5, 57]. Pre-constructed structures provide a fixed high-order relational basis throughout representation learning. Joint structure learning updates high-order relations with graph representations, enabling task-adaptive structural modeling. Thus, scalable integration of high-order modeling and adaptive graph structure learning requires efficient joint optimization.

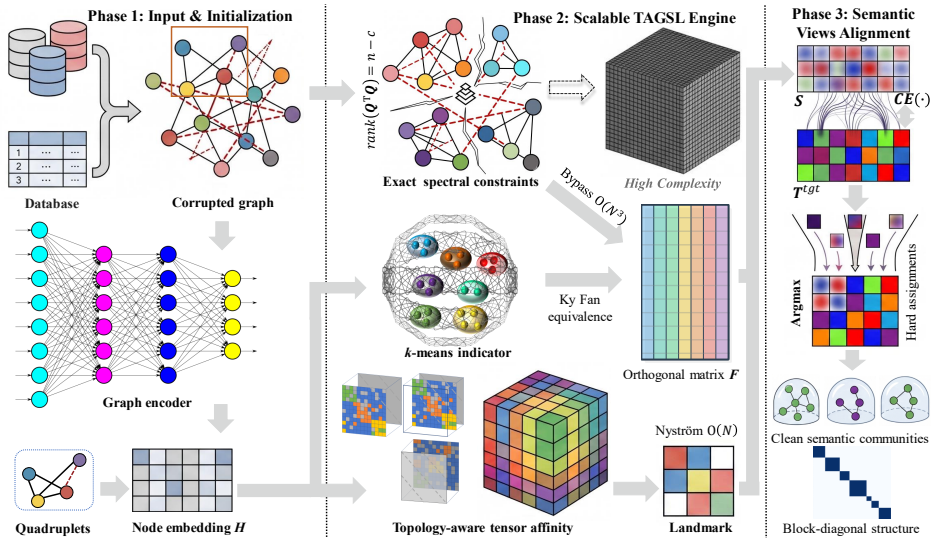


Fig. 2. Overall architecture of TAGSL. **Phase 1** integrates multi-source databases and relational tables to construct an observed graph that may contain spurious linkages. **Phase 2** constructs a topology-aware tensor affinity to refine the graph structure under exact spectral constraints, followed by scalable optimization components to bypass *high complexity*. **Phase 3** aligns the refined structure with semantic distributions to materialize semantic communities with a block-diagonal structure.

3 The Proposed Method

Fig. 2 provides an overview of TAGSL. Section 3.1 formulates unsupervised DGC under structural corruption. Section 3.2 develops topology-aware tensor structure learning and semantic alignment, while Section 3.3 details scalable alternating optimization.

3.1 Problem Formulation

We consider unsupervised DGC on an observed attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{X})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the node set and observed edge set, respectively. The graph contains $n = |\mathcal{V}|$ nodes. An adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ encodes the observed topology, and a feature matrix $\boldsymbol{X} \in \mathbb{R}^{n\times d'}$ represents the node attributes. Following common DGC evaluation practice, the target community count c is assumed known, while node labels remain unavailable during training. Graph databases constructed from multi-source records or relational tables may contain unknown spurious linkages in $\boldsymbol{A}$ [22, 39], which can obscure the latent community structure.

The learning targets comprise an adaptive structural operator $\boldsymbol{L} \in \mathbb{R}^{n\times n}$, a node embedding matrix $\boldsymbol{H} \in \mathbb{R}^{n\times d}$, and a soft community assignment matrix $\boldsymbol{S} \in \mathbb{R}^{n\times c}$. Each entry s_{ij} measures the assignment strength of node i to community j, and the discrete label is obtained as $y_i = \arg\max_j s_{ij}$. The learned structure supports the formation of node embeddings and community assignments that capture c semantically coherent communities. The task is to jointly infer $(\boldsymbol{L}, \boldsymbol{H}, \boldsymbol{S})$ from $(\boldsymbol{A}, \boldsymbol{X}, c)$ without node labels or known locations of spurious linkages. Table 1 summarizes the key notation used in the TAGSL framework. Method-specific symbols are defined where they first appear.

To address the formulated problem, our framework comprises **Phase 1 (Input & Initialization)**, **Phase 2 (Scalable TAGSL Engine)**, and **Phase 3 (Semantic Alignment)**, as illustrated in **Fig. 2**. Specifically, **Phase 1** integrates multi-source databases and relational tables, a process that may introduce spurious linkages and yield a corrupted graph. The **TAGSL** framework explicitly couples

Table 1. Key notation used in the TAGSL framework.

Symbol	Description
$\mathcal{G}$	Observed attributed graph $(\mathcal{V}, \mathcal{E}, \boldsymbol{X})$
n and c	Number of nodes and target communities
$\boldsymbol{A}$	Observed adjacency matrix in $\mathbb{R}^{n\times n}$
$\boldsymbol{X}$	Node feature matrix in $\mathbb{R}^{n\times d'}$
$\boldsymbol{L}$	Adaptive structural operator in $\mathbb{R}^{n\times n}$
$\boldsymbol{H}$	Latent node embedding matrix in $\mathbb{R}^{n\times d}$
$\boldsymbol{S}$	Soft community assignment matrix in $\mathbb{R}^{n\times c}$
y_i	Discrete community label of node i
d' and d	Input feature and latent embedding dimensions

high-order interaction modeling with adaptive graph structure learning. Section 3.2 constructs the topological learning module (**Phase 2**) to extract a refined topological structure. TAGSL derives a topology-aware tensor affinity to reduce the influence of spurious linkages during high-order interaction modeling. The framework further introduces an adaptive structural operator under exact spectral constraints. The prescribed rank condition fixes the null-space dimension at the target community count, supporting exact graph structural decoupling. The semantic alignment stage (**Phase 3**) refines semantic clustering distributions to generate hard community assignments. The resulting assignments organize nodes into semantic communities and provide a block-diagonal representation of the community partition. Section 3.3 details the scalable alternating optimization strategy for **Phase 2**. The Ky Fan reformulation enables tractable optimization of the rank constraint through an auxiliary orthogonal matrix. The separable Nyström approximation reduces the computational burden of tensor operations.

3.2 Topology-Aware Tensor Structure Learning

A topology-aware tensor affinity is first constructed to model multi-node interactions. The structural operator then adapts to the evolving embedding geometry under exact spectral constraints. Finally, semantic alignment maps the learned embeddings to community assignments.

Topology-Aware Tensor Affinity. High-order tensor affinities have shown potential for modeling multi-node interactions [7]. Existing designs utilizing fixed bandwidths fail to adapt to the topological densities inherent in complex graphs. Under structural corruption, fixed-scale or feature-based affinities may inadequately reflect variations in local topological support. We introduce topology-aware modulation weights for each quadruplet (i, j, k, l):

$$w^{(n)}_{ijkl} = (\boldsymbol{A}_{ij} + [\boldsymbol{A}^2]_{ij})(\boldsymbol{A}_{kl} + [\boldsymbol{A}^2]_{kl}), \quad w^{(d)}_{ijkl} = (\boldsymbol{A}_{ik} + [\boldsymbol{A}^2]_{ik}) + (\boldsymbol{A}_{jl} + [\boldsymbol{A}^2]_{jl}), \tag{1}$$

where (i, j) and (k, l) are the target node pairs, (i, k) and (j, l) form the cross-pair context, and $[\boldsymbol{A}^2]_{ij}$ quantifies the two-hop connectivity among nodes i and j.

Restricting connectivity to 2-hop balances contextual richness with efficiency, while symmetric normalization of $\boldsymbol{A}$ and $\boldsymbol{A}^2$ avoids degree explosion and instability. The multiplicative $w^{(n)}_{ijkl}$ captures joint topological support for the target node pairs, whereas the additive $w^{(d)}_{ijkl}$ aggregates cross-pair connectivity to characterize the local structural background. The modulation weight $w^{(n)}_{ijkl}$ integrates target-pair topological support into the feature divergence:

$$\mathcal{E}_{\text{int}} = w^{(n)}_{ijkl}(d_{ij} + d_{kl}), \tag{2}$$

where $d_{ij} = \|\boldsymbol{H}_{i,:} - \boldsymbol{H}_{j,:}\|_2^2$ denotes the squared Euclidean distance.

The local context scale combines cross-pair connectivity and distances to characterize the surrounding structural background:

$$\mathcal{E}_{\text{ext}} = w_{ijkl}^{(d)}(d_{ik} + d_{jl}) + \varepsilon, \tag{3}$$

with $\varepsilon > 0$ ensuring numerical stability.

The Maximum Entropy Principle [54] yields an adaptive affinity distribution under heterogeneous structural densities. Under local consistency constraints, the optimal metric $\boldsymbol{\mathcal{T}}$ takes an exponential decay form with the topological Signal-to-Background Ratio:

$$\boldsymbol{\mathcal{T}}_{ijkl} = 1 - \exp\left(-\sigma \frac{\mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right), \tag{4}$$

where $\sigma > 0$ is a constant. The ratio $\mathcal{E}_{\text{int}}/\mathcal{E}_{\text{ext}}$ normalizes target-pair divergence against local structural background. Intuitively, affinity strength for the same target-pair divergence varies with local structural conditions. Theorem 1 formalizes the coupling between target-pair divergence and cross-pair context.

Theorem 1. *The tensor metric $\boldsymbol{\mathcal{T}}$ enforces an irreducible non-separable coupling. The local context scale $\mathcal{E}_{ext}$ modulates its sensitivity to the internal divergence $\mathcal{E}_{int}$, yielding a generally non-zero mixed partial derivative $\frac{\partial^2 \boldsymbol{\mathcal{T}}}{\partial \mathcal{E}_{int} \partial \mathcal{E}_{ext}}$.*

The mixed partial derivative shows that local context alters the sensitivity of $\boldsymbol{\mathcal{T}}$ to internal divergence, allowing local topology to modulate target-pair interaction strength. Complete theoretical proofs are provided in the appendix. The tensor affinity integrates local structural context into high-order interactions for subsequent adaptive graph structure learning.

Adaptive Graph Structure Learning. A standard GNN objective balances feature reconstruction with pairwise Laplacian smoothness. Given the initial graph node feature matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d'}$ and adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, the static Laplacian operator is defined as $\boldsymbol{L}^{(0)} = \boldsymbol{I} - \boldsymbol{A}$. The node embedding $\boldsymbol{H} \in \mathbb{R}^{n \times d}$ is learned via the projection $\boldsymbol{W} \in \mathbb{R}^{d' \times d}$ by minimizing the graph function:

$$\boldsymbol{\mathcal{L}}(\boldsymbol{H}) = \|\boldsymbol{H} - \boldsymbol{X}\boldsymbol{W}\|_F^2 + \alpha \operatorname{Tr}(\boldsymbol{H}^\top \boldsymbol{L}^{(0)} \boldsymbol{H}), \tag{5}$$

where $\alpha > 0$ balances graph smoothness. Eq. (5) provides local pairwise consistency through a fixed operator derived from the observed adjacency matrix. Under structural corruption, the smoothness term inherits spurious linkages from $\boldsymbol{A}$, motivating structural guidance that evolves with the embedding geometry.

As illustrated in Phase 2 of Fig. 2, TAGSL extends the pairwise formulation by introducing the topology-aware tensor affinity $\boldsymbol{\mathcal{T}}$ and replacing the fixed $\boldsymbol{L}^{(0)}$ with a learnable structural operator $\boldsymbol{L}$. The tensor affinity captures topology-conditioned multi-way interactions, while $\boldsymbol{L}$ adapts structural guidance to the evolving embedding geometry. We parameterize $\boldsymbol{L} = \boldsymbol{Q}^\top \boldsymbol{Q}$ with a learnable spectral factor $\boldsymbol{Q} \in \mathbb{R}^{(n-c) \times n}$, which ensures a positive semidefinite structural energy and enables the rank constraint with the target community count c. The resulting objective jointly optimizes feature reconstruction, adaptive structural smoothness, and topology-aware high-order interactions:

$$\begin{aligned} \boldsymbol{\mathcal{L}}(\boldsymbol{H}, \boldsymbol{Q}) = \|\boldsymbol{H} - \boldsymbol{X}\boldsymbol{W}\|_F^2 + \alpha \operatorname{Tr}(\boldsymbol{H}^\top \boldsymbol{Q}^\top \boldsymbol{Q} \boldsymbol{H}) + \beta \sum_{s=1}^{d} \boldsymbol{\mathcal{T}} \times_4 \boldsymbol{H}_{:,s} \times_3 \boldsymbol{H}_{:,s} \times_2 \boldsymbol{H}_{:,s} \times_1 \boldsymbol{H}_{:,s}. \\ \text{s.t.} \quad \boldsymbol{H}^\top \boldsymbol{H} = \boldsymbol{I}, \operatorname{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c, \boldsymbol{Q}\boldsymbol{1} = \boldsymbol{0}, \end{aligned} \tag{6}$$

where $\beta > 0$ controls the topology-aware tensor affinity term modeling multi-way interactions among graph quadruples, and $\times_k$ ($k \in \{1, 2, 3, 4\}$) denotes the mode-k tensor product.

Rationale for Structural Design. Eq. (6) jointly regulates the embedding geometry and the learned structural operator. For the embedding geometry, the orthogonality constraint $\boldsymbol{H}^\top \boldsymbol{H} = \boldsymbol{I}$ prevents representational collapse and maintains orthogonal latent dimensions. For the structural operator, the factorization $\boldsymbol{L} = \boldsymbol{Q}^\top \boldsymbol{Q}$ guarantees a positive semidefinite energy. The constraint $\boldsymbol{Q}\mathbf{1} = 0$ places the constant vector in the null space of $\boldsymbol{L}$, while $\operatorname{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c$ fixes the null space dimension at c and aligns the spectral structure with the target community count. This parameterization allows smoothness to act on a learned $\boldsymbol{L}$ that evolves with the embedding geometry. The tensor affinity supplies complementary local one-hop and two-hop structural context, providing a unified formulation for graphs with varying homophily ratios. Scale normalization of $\boldsymbol{Q}$ controls the factor magnitude and prevents trivial scale collapse during optimization.

Semantic Alignment and Joint Objective. Structural optimization produces the continuous embedding $\boldsymbol{H}$, while Phase 3 of Fig. 2 estimates the explicit community assignments required by the problem formulation. A self-supervised clustering module [46] converts distances to learnable cluster centroids into soft assignments through a Student's t-distribution:

$$s_{ij} = \frac{(1 + \|\boldsymbol{H}_{i,:} - \boldsymbol{\mu}_j\|_2^2)^{-1}}{\sum_{j'} (1 + \|\boldsymbol{H}_{i,:} - \boldsymbol{\mu}_{j'}\|_2^2)^{-1}}, \tag{7}$$

where $\boldsymbol{\mu}_j$ represents the j-th learnable cluster centroid, typically initialized via K-means over the node embeddings.

The auxiliary clustering loss combines consistency with the observed topology and self-supervised assignment refinement:

$$\boldsymbol{\mathcal{L}}_{\text{cluster}} = \text{MSE}(\boldsymbol{H}\boldsymbol{H}^\top, \boldsymbol{A}) + \text{CE}(\boldsymbol{T}^{\text{tgt}}, \boldsymbol{S}), \tag{8}$$

where MSE measures consistency between $\boldsymbol{H}\boldsymbol{H}^\top$ and the observed adjacency matrix $\boldsymbol{A}$, and CE aligns $\boldsymbol{S}$ with the self-supervised target distribution $\boldsymbol{T}^{\text{tgt}}$. The target distribution is constructed from $\boldsymbol{S}$ without node labels and emphasizes high-confidence assignments:

$$t_{ij}^{\text{tgt}} = \frac{s_{ij}^2 / \sum_i s_{ij}}{\sum_{j'} s_{ij'}^2 / \sum_i s_{ij'}}. \tag{9}$$

The high-order structural energy and semantic alignment loss form the unified TAGSL objective:

$$\mathcal{J} = \mathcal{L}(\boldsymbol{H}, \boldsymbol{Q}) + \boldsymbol{\mathcal{L}}_{\text{cluster}}. \tag{10}$$

The discrete label of node i is obtained as $y_i = \arg\max_j s_{ij}$.

3.3 Scalable Optimization Strategies

The network parameters $\boldsymbol{W}$ and cluster centroids $\boldsymbol{\mu}$ are updated through standard backpropagation on the joint objective $\mathcal{J}$. The structural variables require specialized updates because $\boldsymbol{Q}$ is subject to exact spectral constraints, while $\boldsymbol{H}$ is constrained by orthogonality and a computationally intensive fourth-order tensor term. Accordingly, TAGSL alternates between the constrained updates of $\boldsymbol{Q}$ and $\boldsymbol{H}$, followed by the semantic network update.

Tractable Optimization of Q under Exact Spectral Constraints. With the node embeddings $\boldsymbol{H}$ fixed, $\mathcal{L}(\boldsymbol{H}, \boldsymbol{Q})$ reduces to optimizing the spectral factor $\boldsymbol{Q}$ under the exact rank constraint $\operatorname{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c$ and the zero-row-sum constraint $\boldsymbol{Q}\mathbf{1} = \mathbf{0}$ as follows:

$$\begin{aligned} \min_{\boldsymbol{Q}} \quad & \alpha \operatorname{Tr}(\boldsymbol{H}^\top \boldsymbol{Q}^\top \boldsymbol{Q} \boldsymbol{H}) \\ \text{s.t.} \quad & \operatorname{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c, \quad \boldsymbol{Q}\mathbf{1} = \mathbf{0}. \end{aligned} \tag{11}$$

Direct enforcement of the exact rank constraint is difficult in gradient optimization. Following Ky Fan's theorem [11], the rank requirement is reformulated as the null-space $\operatorname{Tr}(\boldsymbol{F}^\top \boldsymbol{Q}^\top \boldsymbol{Q} \boldsymbol{F}) = 0$, where $\boldsymbol{F} \in \mathbb{R}^{n \times c}$ is orthonormal. The matrix $\boldsymbol{F}$ is constructed by orthonormalizing the indicator matrix obtained from K-means clustering on the current embeddings $\boldsymbol{H}$. The identity $\operatorname{Tr}(\boldsymbol{F}^\top \boldsymbol{Q}^\top \boldsymbol{Q} \boldsymbol{F}) = \|\boldsymbol{Q}\boldsymbol{F}\|_F^2$ converts the spectral condition into a Frobenius-norm constraint that can be incorporated into the augmented Lagrangian.

Row-wise mean-centering is applied after each gradient update to directly enforce the zero-row-sum constraint $\boldsymbol{Q}\mathbf{1} = \mathbf{0}$. The centering step requires no additional penalty coefficient, while the augmented Lagrangian handles the spectral null-space condition $\|\boldsymbol{Q}\boldsymbol{F}\|_F^2 = 0$. Combining the Ky Fan reformulation with row-wise mean-centering yields the following objective for updating $\boldsymbol{Q}$:

$$\mathcal{L}(\boldsymbol{Q}) = \alpha \|\boldsymbol{Q}\boldsymbol{H}\|_F^2 + \lambda \|\boldsymbol{Q}\boldsymbol{F}\|_F^2 + \frac{\rho}{2} \|\boldsymbol{Q}\boldsymbol{F}\|_F^4, \tag{12}$$

where λ is the dual variable and $\rho > 0$ is the penalty coefficient. The quartic penalty increases the cost of large violations of the spectral null-space condition while preserving differentiability. The gradient update for $\boldsymbol{Q}$ is:

$$\boldsymbol{Q} \leftarrow \boldsymbol{Q} - 2\eta \Big(\alpha \boldsymbol{Q}\boldsymbol{H}\boldsymbol{H}^\top + \lambda \boldsymbol{Q}\boldsymbol{F}\boldsymbol{F}^\top + \rho \|\boldsymbol{Q}\boldsymbol{F}\|_F^2 \boldsymbol{Q}\boldsymbol{F}\boldsymbol{F}^\top \Big), \tag{13}$$

where $\eta > 0$ denotes the learning rate. After the primal update, the dual variable is updated as $\lambda = \lambda + \rho \|\boldsymbol{Q}\boldsymbol{F}\|_F^2$, and row-wise mean-centering restores the zero-row-sum constraint. The Ky Fan reformulation and centering operation avoid an explicit eigendecomposition during each $\boldsymbol{Q}$-update and separate the treatment of the spectral and zero-row-sum constraints.

Scalable Optimization of $\boldsymbol{H}$ via Nyström Approximation. Fixing $\boldsymbol{Q}$, $\mathcal{L}(\boldsymbol{H}, \boldsymbol{Q})$ with respect to $\boldsymbol{H}$ reduces to:

$$\begin{aligned} \min_{\boldsymbol{H}} \quad & \|\boldsymbol{H} - \boldsymbol{X}\boldsymbol{W}\|_F^2 + \alpha \|\boldsymbol{Q}\boldsymbol{H}\|_F^2 + \beta \sum_{s=1}^{d} \boldsymbol{\mathcal{T}} \times_4 \boldsymbol{H}_{:,s} \times_3 \boldsymbol{H}_{:,s} \times_2 \boldsymbol{H}_{:,s} \times_1 \boldsymbol{H}_{:,s}, \\ & \text{s.t.} \quad \boldsymbol{H}^\top \boldsymbol{H} = \boldsymbol{I}. \end{aligned} \tag{14}$$

Computational cost is dominated by the fourth-order tensor contraction, whose unfolded form $\boldsymbol{\mathcal{T}}_{(1)} \in \mathbb{R}^{n^2 \times n^2}$ grows rapidly with graph size. To make the contraction tractable, we approximate the unfolded tensor through the low-rank factorization $\boldsymbol{\mathcal{T}}_{(1)} \approx \boldsymbol{P}^\top \boldsymbol{P}$, where $\boldsymbol{P} \in \mathbb{R}^{r \times n^2}$ and $r \ll n^2$. Under this factorization, the contraction is written as:

$$\sum_{s=1}^{d} \boldsymbol{\mathcal{T}} \times_4 \boldsymbol{H}_{:,s} \times_3 \boldsymbol{H}_{:,s} \times_2 \boldsymbol{H}_{:,s} \times_1 \boldsymbol{H}_{:,s} = \operatorname{Tr}(\boldsymbol{B}^\top \boldsymbol{\mathcal{T}}_{(1)} \boldsymbol{B}) \approx \|\boldsymbol{P}\boldsymbol{B}\|_F^2, \tag{15}$$

where $\boldsymbol{B} = \boldsymbol{H} * \boldsymbol{H} \in \mathbb{R}^{n^2 \times d}$ and $*$ denotes the Khatri-Rao product. Low-rank factorization reduces tensor construction to operations with $\boldsymbol{P}$. We further reduce computation over the n^2-dimensional representation using the separable Nyström form $\boldsymbol{P} \cong \boldsymbol{P}_n \otimes \boldsymbol{P}_n$ [45], with $\boldsymbol{P}_n \in \mathbb{R}^{m \times n}$ and $m \ll n$. The mixed-product property of Khatri-Rao and Kronecker products yields:

$$\boldsymbol{P}\boldsymbol{B} = (\boldsymbol{P}_n \otimes \boldsymbol{P}_n)(\boldsymbol{H} * \boldsymbol{H}) = (\boldsymbol{P}_n \boldsymbol{H}) * (\boldsymbol{P}_n \boldsymbol{H}). \tag{16}$$

Algorithm 1 Training Procedure of TAGSL

Require: Input node features $\boldsymbol{X}$, initial adjacency $\boldsymbol{A}$, number of clusters c, learning rate η, structural weights α, β

Initialize: Network weights $\boldsymbol{W}$, embeddings $\boldsymbol{H}$, spectral factor $\boldsymbol{Q}$, cluster centroids $\boldsymbol{\mu}$, dual variable $\lambda = 0$, penalty $\rho = 1$
Construct: Landmark affinity matrix $\boldsymbol{P}_n$ for separable Nyström approximation
for each training iteration t **do**
  % 1. Tractable Optimization of $\boldsymbol{Q}$
  Construct orthonormal $\boldsymbol{F}$ based on current $\boldsymbol{H}$ via K-means
  Update $\boldsymbol{Q}$ via Eq. (13)
  Apply row-wise mean-centering on $\boldsymbol{Q}$ to satisfy $\boldsymbol{Q}\mathbf{1} = \mathbf{0}$
  Update dual variable $\lambda = \lambda + \rho\|\boldsymbol{Q}\boldsymbol{F}\|_F^2$
  % 2. Scalable Optimization of $\boldsymbol{H}$
  Compute structural gradients $\nabla_{\boldsymbol{H}}\mathcal{L}$ via Eq. (18)
  Tentative update: $\boldsymbol{H}^{(t+1/2)} = \boldsymbol{H}^{(t)} - \eta\nabla_{\boldsymbol{H}}\mathcal{L}$
  Apply decomposition $\boldsymbol{H}^{(t+1/2)} = \boldsymbol{U}\boldsymbol{R}$ and set $\boldsymbol{H}^{(t+1)} = \boldsymbol{U}$
  % 3. Semantic Alignment and Network Update
  Compute soft assignments $\boldsymbol{S}$ and target distribution $\boldsymbol{T}^{\text{tgt}}$
  Update $\boldsymbol{W}$ and $\boldsymbol{\mu}$ by minimizing total loss $\mathcal{J}$
end for
Postprocessing: Determine discrete labels $y_i = \arg\max_j s_{ij}$
return Embeddings $\boldsymbol{H}$, predicted cluster labels $\boldsymbol{y}$

Let $\boldsymbol{u}_k$ denote the k-th column of $\boldsymbol{P}_n\boldsymbol{H}$. Taking the Frobenius norm yields:

$$\|\boldsymbol{P}\boldsymbol{B}\|_F^2 = \sum_{k=1}^{d} \|\boldsymbol{u}_k \otimes \boldsymbol{u}_k\|_2^2 = \sum_{k=1}^{d} \|\boldsymbol{u}_k\|_2^4. \tag{17}$$

The Nyström factorization produces a quartic expression in the projected embeddings. For a smoother and lower-cost update, TAGSL optimizes the quadratic surrogate $\sum_{k=1}^{d} \|\boldsymbol{u}_k\|_2^2 = \|\boldsymbol{P}_n\boldsymbol{H}\|_F^2$. The resulting partial gradient with respect to $\boldsymbol{H}$ is:

$$\nabla_{\boldsymbol{H}}\mathcal{L} = 2(\boldsymbol{H} - \boldsymbol{X}\boldsymbol{W}) + 2\alpha\boldsymbol{Q}^\top\boldsymbol{Q}\boldsymbol{H} + 2\beta\boldsymbol{P}_n^\top\boldsymbol{P}_n\boldsymbol{H}. \tag{18}$$

The orthogonality constraint is restored after each gradient step through a QR retraction onto the Stiefel manifold. Given $\boldsymbol{H}^{(t+1/2)} = \boldsymbol{H}^{(t)} - \eta\nabla_{\boldsymbol{H}}\mathcal{L}$, we compute $\boldsymbol{H}^{(t+1/2)} = \boldsymbol{U}\boldsymbol{R}$ and set $\boldsymbol{H}^{(t+1)} = \boldsymbol{U}$. The retracted iterate satisfies $(\boldsymbol{H}^{(t+1)})^\top\boldsymbol{H}^{(t+1)} = \boldsymbol{I}$.

Algorithm Summary and Complexity. The alternating optimization procedure is summarized in Algorithm 1. Network weights $\boldsymbol{W}$ and cluster centroids $\boldsymbol{\mu}$ are updated using Adam via backpropagation on the joint objective $\mathcal{J}$. Structural variables $\boldsymbol{Q}$ and $\boldsymbol{H}$ are alternately refined using the derived update rules. Assuming bounded structural iterates, Lipschitz continuous block gradients, and sufficiently small learning rates, the alternating updates of $\boldsymbol{Q}$ and $\boldsymbol{H}$ satisfy sufficient decrease for each block. Under the relative error condition, the lower boundedness and Kurdyka–Łojasiewicz property [4] of $\mathcal{L}(\boldsymbol{H}, \boldsymbol{Q})$ ensure convergence of the structural sequence to a critical point.

The main computational cost arises from evaluating Eq. (18). Sparse graph propagation requires $O(|E|d)$ operations. The structural terms in Eq. (18) are evaluated through low-rank operations with $O(mNd)$ complexity for $m \ll N$. The fourth-order tensor contribution is already included in the $O(mNd)$ term through the separable Nyström approximation. The variable updates take

Table 2. Statistics of Evaluation Graphs

Dataset	Nodes	Edges	Features	Classes	HR
Texas	183	325	1703	5	0.0599
Squirrel	5201	108536	2089	5	0.2221
Flickr	7575	239738	12047	9	0.2386
Cora	2708	5429	1433	7	0.8100
Citeseer	3327	4732	3703	6	0.7355
Pubmed	19717	88651	500	3	0.8024
HPPI	40887	1002714	697	7	0.9997

$O(mN)$ for the row-wise mean-centering of $\boldsymbol{Q}$ and $O(Nd^2)$ for the Thin-QR manifold retraction of $\boldsymbol{H}$. Consequently, the overall time complexity is $O(N(md + d^2) + |E|d)$. For fixed m and d on sparse graphs, the overall complexity grows linearly with the node count N.

4 Experiment

Section 4.1 details the experimental configurations, while Section 4.2 presents the quantitative clustering results and embedding visualizations. Section 4.3 conducts ablation studies to verify the necessity of core structural components, whereas Section 4.4 and Section 4.5 provide analyses of structural coherence and cluster quality. Section 4.6 investigates the sensitivity of structural hyper-parameters, and Section 4.7 validates the computational efficiency. Finally, Section 4.8 demonstrates the practical utility of our framework in a real-world multi-omics database cleaning scenario.

4.1 Experimental Setting

This section summarizes the datasets, baselines, evaluation metrics, and implementation details.

Datasets and Environments. All experiments are conducted using PyTorch Geometric (v2.2.0) on a 64-bit Windows 10 OS with an Intel Core i9-14900 CPU, 64 GB RAM, and the NVIDIA RTX 3090 GPU. We evaluate on seven benchmark graphs, including three heterophilic graphs (Texas, Squirrel [33], Flickr [26]), three homophilic citation networks (Cora, Citeseer, Pubmed [3]), and a real-world protein-protein interaction graph (HPPI). The HPPI graph is constructed from the BioGRID database [1], providing interaction records but no structured graph data. Proteins are mapped to node indices with edges defined by genetic interactions. Node features are generated using BioBERT [21], whose pretraining on biomedical literature provides domain-aligned representations for the HPPI graph. Labels for proteins are assigned via majority voting, considering the most frequent class label across different experimental sources. As public datasets may not sufficiently reflect complex biological or high-order dependencies, HPPI is included to evaluate performance in real-world applications.

Baselines and Evaluation Metrics. Table 2 summarizes the statistical information of evaluation graphs, including the number of nodes, edges, features, classes, and homophily ratios (HR). We evaluate clustering performance using accuracy (ACC) and normalized mutual information (NMI). All experiments are independently conducted 5 times with different random seeds, and the average metrics are reported. Our framework is compared with: **Structural Clustering Methods:** GC-SEE [9] (PR'23), UGT [17] (AAAI'24), DGCN [32] (ICML'23), and UDGC [19] (TNNLS'25), which enhance structure topology-guided learning. **Contrastive Learning Models:** HSAN [28] (AAAI'23), GLAC-GCN [52] (TAI'24), DCLN [35] (TNNLS'24), and GCC [50] (ESA'23), which leverage contrastive

[1]https://downloads.thebiogrid.org/BioGRID/Release-Archive

Table 3. Clustering performance averaged over five independent runs. The best results are highlighted in green, and the second-best results in lightgreen.

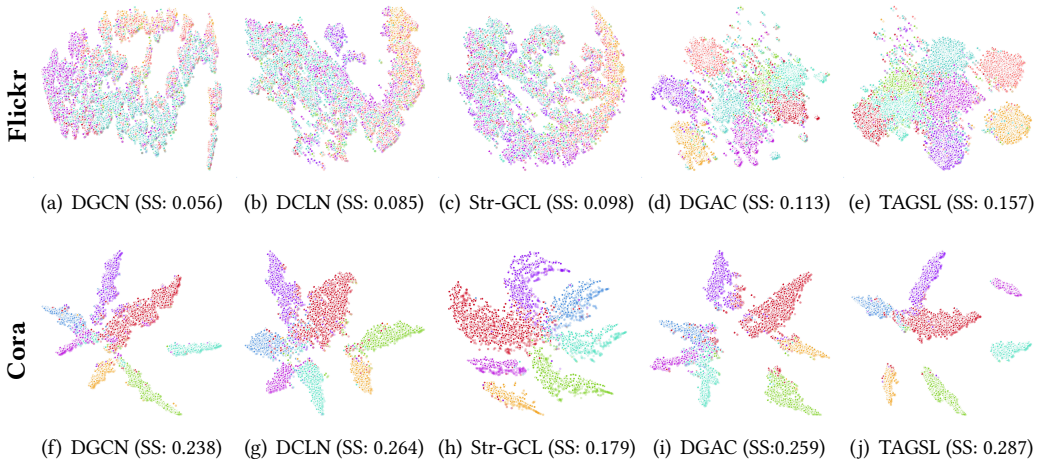
Method	Texas		Squirrel		Flickr		Cora		Citeseer		Pubmed		HPPI		Avg.Rank
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	
GC-SEE	55.19	4.32	25.34	2.75	21.20	12.29	73.58	53.02	70.90	44.00	63.91	24.50	61.32	52.23	9.71
UGT	58.29	15.15	29.26	2.27	27.76	2.95	60.09	46.96	69.76	44.02	60.54	22.92	70.65	36.79	9.00
DGCN	74.57	39.93	31.39	8.54	28.55	16.21	72.89	56.82	71.60	44.83	53.70	15.04	67.36	38.40	6.07
UDGC	59.02	17.76	25.81	3.34	29.74	17.02	76.20	60.20	70.75	43.78	61.18	33.37	69.13	37.21	6.79
HSAN	60.38	15.09	28.69	3.83	30.94	18.60	77.07	59.21	71.15	45.06	60.56	26.24	59.60	26.88	6.61
GLAC-GCN	58.20	14.75	25.92	3.93	28.19	17.95	62.81	39.22	71.12	44.50	54.83	14.59	49.95	24.32	9.93
DCLN	60.38	15.24	28.58	3.68	30.71	18.78	77.30	**62.24**	70.02	46.40	68.32	32.20	63.53	27.10	5.46
GCC	61.50	21.48	27.23	2.98	27.68	17.71	75.15	56.90	69.31	43.61	67.43	33.07	61.14	26.67	8.14
ASHGCL	58.29	15.15	25.90	3.92	26.12	17.77	68.71	51.82	64.83	40.13	44.08	11.57	61.28	27.73	10.07
Str-GCL	60.22	14.77	28.55	3.74	31.22	18.99	56.56	50.67	41.77	43.49	38.36	10.42	58.92	26.34	10.07
FPGC	51.26	14.39	25.68	4.26	19.16	6.27	79.19	59.55	72.59	46.36	71.03	34.57	64.48	31.42	6.86
ADIIN	56.17	7.03	28.70	1.73	24.54	12.53	52.24	41.79	71.87	45.18	67.73	30.50	57.12	28.33	9.43
DGAC	75.08	46.19	34.43	**12.24**	81.59	66.36	75.91	56.18	69.59	43.77	70.82	34.26	55.09	23.12	5.71
TAGSL (Ours)	**77.04**	**47.66**	**35.33**	12.14	**84.98**	**71.05**	**79.87**	60.90	**73.54**	**47.29**	**72.53**	**36.75**	**71.59**	**55.84**	**1.14**

objectives with graph augmentations. **Structure-Aware Contrastive Methods:** ASHGCL [18] (Inf. Fusion'25) and Str-GCL [16] (WWW'25), which integrate structural priors or motifs into contrastive frameworks. **High-Order Topology Models:** FPGC [51] (AAAI'25), ADIIN [24] (KBS'25), and DGAC [49] (WWW'25), which explicitly model complex dependencies beyond pairwise interactions. For fair comparison, all baselines are evaluated using official implementations or faithful reproductions, with hyperparameters set according to original papers.

Implementation Details. The dimension of node embeddings is fixed at $d = 256$, and the model is trained for a maximum of 200 epochs. The projection matrix $\boldsymbol{W}$ and cluster centroids $\boldsymbol{\mu}$ are optimized using the Adam optimizer. The update step size η in Eq. (13) shares the unified base learning rate of 0.001. The Nyström landmark size is fixed at $m = 256$. For each dataset, the core structural weights α and β are selected from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$ by maximizing the Silhouette Score [38] computed on the learned embeddings using the predicted cluster assignments. Ground-truth labels are reserved for computing the reported ACC and NMI.

4.2 Clustering Performance and Visualization

Our comparison baselines span from pairwise structural modeling to high-order inference, and the quantitative mean results are shown in Table 3. The Avg. Rank column denotes the mean rank across all combinations (lower is better). On heterophilic graphs, ours achieves superior results. On Texas, it obtains 77.04% ACC and 47.66% NMI, outperforming the second-best DGAC (75.08%, 46.19%). On Squirrel, it reaches 35.33% ACC and 12.14% NMI, surpassing DGAC in ACC while achieving a similar NMI (12.24%). On Flickr, ours achieves 84.98% ACC and 71.05% NMI, exceeding DGAC (81.59%, 66.36%). On homophilic graphs, ours remains competitive. On Cora, it attains 79.87% ACC and 60.90% NMI, surpassing FPGC (79.19% ACC and 59.55% NMI) and performing comparably to DCLN. On Citeseer, it yields 73.54% ACC and 47.29% NMI, outperforming FPGC (72.59%, 46.36%). On Pubmed, ours achieves 72.53% ACC and 36.75% NMI, surpassing FPGC (71.03%, 34.57%). On HPPI, ours attains 71.59% ACC and 55.84% NMI, outperforming UGT in ACC (70.65%), GC-SEE in NMI



(a) DGCN (SS: 0.056) (b) DCLN (SS: 0.085) (c) Str-GCL (SS: 0.098) (d) DGAC (SS: 0.113) (e) TAGSL (SS: 0.157)

(f) DGCN (SS: 0.238) (g) DCLN (SS: 0.264) (h) Str-GCL (SS: 0.179) (i) DGAC (SS:0.259) (j) TAGSL (SS: 0.287)

Fig. 3. The t-SNE visualization of node embeddings on Flickr and Cora, with colors denoting ground-truth classes. Silhouette Scores (SS), computed on the original embeddings using predicted assignments, provide a quantitative complement. TAGSL shows relatively clear cluster boundaries and reduced class overlap, consistent with the highest SS on both datasets.

(52.23%), and FPGC on both metrics. The low homophily ratio of Flickr limits the class-consistent evidence available from direct neighborhoods. The strong performance of DGAC and TAGSL is consistent with their use of broader structural context: DGAC models high-order semantics through graph diffusion, whereas TAGSL combines topology-aware multi-node modulation with a learned structural operator. The strong performance of TAGSL across diverse graph regimes supports its effectiveness under varying structural conditions. On challenging datasets such as Squirrel and HPPI, heterophily or structural noise can reduce the reliability of static pairwise propagation. TAGSL incorporates local structural densities into multi-node interaction modeling to reduce the influence of potentially spurious relations. Results on Cora and Pubmed further indicate that the high-order refinement remains compatible with informative homophilic structure. Across these regimes, the results are consistent with the complementary roles of topology-aware tensor modulation and exact spectral constraints.

To assess the embeddings, t-SNE visualizations are provided in Fig. 3. On Flickr, DGCN, DCLN, and Str-GCL (Fig. 3(a)–(c)) show substantial overlap among ground-truth classes. DGAC (d) forms several partially separated regions. TAGSL (e) shows clearer grouping among several classes, although class mixing remains in the central region. On Cora, the baseline embeddings (f)–(i) retain varying degrees of overlap. TAGSL (j) shows clearer separation among several ground-truth classes and more coherent class regions. The colors in the t-SNE plots denote ground-truth classes. The SS values are computed on the original embeddings using predicted cluster assignments and provide an unsupervised quantitative complement. On Flickr, TAGSL achieves an SS of 0.157, compared with 0.113 for DGAC. On Cora, TAGSL achieves an SS of 0.287, compared with 0.264 for DCLN. The visual patterns and SS values indicate improved embedding separability on both graphs.

Overall, Table 3 shows strong clustering performance for TAGSL. Across the seven datasets and two evaluation metrics, TAGSL ranks first in 12 cases and second in 2 cases. The resulting average rank is 1.14. Fig. 3 complements the quantitative results with visual evidence of reduced class overlap. TAGSL also achieves the highest SS among the compared methods on both Flickr and Cora. The

results across heterophilic, homophilic, and biological graphs support the effectiveness of combining topology-aware multi-node modeling with adaptive graph structure learning. Subsequent ablation studies examine the contribution of each structural component.

Table 4. Ablation study of the main TAGSL components. T denotes the topology-aware tensor affinity, O denotes the embedding orthogonality constraint, and L denotes the learned structural operator under exact spectral constraints. The best results are shown in bold, and the second-best results are underlined.

T	O	L	Texas		Squirrel		Flickr		Cora		Citeseer		Pubmed		HPPI		Avg.Rank
			ACC.	NMI	ACC.	NMI	ACC.	NMI	ACC.	NMI	ACC.	NMI	ACC.	NMI	ACC.	NMI	
			67.45	38.44	27.27	7.14	69.41	48.83	72.40	51.67	66.62	39.39	65.23	29.17	64.63	35.74	8.00
✔			71.70	42.98	30.54	9.61	73.55	54.20	77.27	56.37	71.33	43.93	69.96	33.65	69.21	40.11	4.00
	✔		68.90	39.79	28.59	8.04	71.06	50.69	73.88	53.71	68.17	40.89	66.87	31.32	66.90	37.45	6.93
		✔	68.17	41.54	29.45	8.56	73.12	52.67	75.82	54.39	70.13	42.56	69.00	32.88	68.23	38.95	6.07
✔	✔		72.38	43.38	30.75	9.99	74.09	54.88	77.15	57.70	71.35	44.76	69.97	34.58	69.07	40.96	3.14
✔		✔	73.71	44.10	31.30	10.30	74.83	55.61	77.81	58.90	72.12	45.62	70.77	35.44	69.78	41.78	2.00
	✔	✔	71.69	42.56	30.31	9.42	73.81	53.75	76.97	56.51	70.89	43.46	69.79	33.39	68.93	39.86	4.86
✔	✔	✔	**77.04**	**47.66**	**35.33**	**12.14**	**84.98**	**71.05**	**79.87**	**60.90**	**73.54**	**47.29**	**72.53**	**36.75**	**71.59**	**55.84**	**1.00**

4.3 Ablation on TAGSL Components

We evaluate the impact of three core components in the proposed **TAGSL**: the topology-aware tensor affinity (T), the embedding orthogonality constraint (O), and the learned structural operator (L) under the exact spectral rank constraint $\text{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c$ and the zero-row-sum constraint $\boldsymbol{Q}\mathbf{1} = \mathbf{0}$. We evaluate all eight combinations of the three components. All variants retain auxiliary regularization for fair comparison. Results are presented in Table 4. The table reports ACC and NMI for all seven datasets, together with the average rank of each variant. The discussion below highlights Flickr as a heterophilic graph and Cora as a homophilic graph, while HPPI represents a biological network. Disabling all components yields the weakest performance on ACC and NMI, with an average rank of 8.00. Each component improves ACC and NMI on every dataset when used alone. The topology-aware tensor affinity T performs best among the three individual configurations for every dataset and metric. The average rank is 4.00 for T, 6.07 for the learned structural operator L, and 6.93 for the orthogonality constraint O. The average gains over the variant without the three components further distinguish the three individual configurations. The topology-aware tensor affinity T improves ACC by 4.36 points and NMI by 4.35 points. The gains from L are 2.98 points in ACC and 3.03 points in NMI. The gains from O are 1.62 points in ACC and 1.65 points in NMI. The improvement from T is consistently observed on Flickr, Cora, and HPPI. This pattern is consistent with the design of T, which incorporates local structural context when modeling interactions among multiple nodes.

The combination of T and L ($T + L$) attains an average rank of 2.00, the lowest among the three configurations containing two components. The average ranks are 3.14 for the combination of T and O ($T + O$) and 4.86 for the combination of O and L ($O + L$). Combining T and L performs best among the three configurations on every dataset and metric. The combination of T and L ranks second only to the complete model for every reported ACC and NMI result. Adding L to T improves every ACC and NMI value. The average gains are 0.97 points in ACC and 1.56 points in NMI. The combination of T and L improves average ACC by 0.80 points and average NMI by 0.79 points over the combination of T and O. On Flickr, Cora, and HPPI, combining T and L yields ACC/NMI values of 74.83%/55.61%, 77.81%/58.90%, and 69.78%/41.78%. The consistent advantage of combining T and L supports complementary roles for the tensor affinity and learned structural operator in TAGSL.

The tensor affinity incorporates local structural context into interactions among multiple nodes, while the learned structural operator guides graph refinement through exact spectral constraints.

The complete model (T+O+L) achieves an average rank of 1.00 and performs best on every dataset and metric. Every reported ACC and NMI value increases after O is added to the configuration containing T and L. The average ACC gain is 3.51 points, and the average NMI gain is 5.70 points. Adding O raises ACC/NMI from 74.83%/55.61% to 84.98%/71.05% on Flickr and from 77.81%/58.90% to 79.87%/60.90% on Cora. On HPPI, ACC/NMI increases from 69.78%/41.78% to 71.59%/55.84%. The NMI gains are especially large on Flickr and HPPI. The configuration containing only O provides relatively modest gains, whereas adding O with T and L produces larger improvements. The larger gains suggest a stronger contribution of the orthogonality constraint in the complete model. The additional gains are consistent with the role of orthogonality in regulating embedding geometry during optimization of the tensor affinity and learned structural operator.

Table 4 shows that the tensor affinity T provides the largest gain when used alone. Combining T with L produces the strongest configuration containing two components, and adding O yields further gains in the complete model. The overall ablation pattern supports complementary roles for T, L, and O in interaction modeling, graph structure learning, and embedding geometry regulation.

4.4 Affinity and Graph Structure Analysis

This section examines topological modulation in the tensor affinity and the organization of similarities in the learned embedding space.

Effect of Topological Modulation in Tensor Affinity. We conduct a controlled comparison to examine the effect of topological modulation. Cora, Flickr, and HPPI represent homophilic, heterophilic, and biological graphs. For each dataset, the sampled subset prioritizes highly connected nodes while retaining randomly selected nodes from the full graph. The baseline model relies on purely feature-based distance affinities, whereas our proposed variant incorporates topology-aware modulation via 1-hop and 2-hop connectivity. For each dataset, both settings use the same sampled nodes and the same 1-hop and 2-hop connection pairs. We define *Connection Clustering Consistency* as the proportion of sampled 1-hop or 2-hop connections whose endpoints are assigned to the same predicted cluster. Table 5 shows improvements in both 1-hop and 2-hop connection consistency across all three sampled subsets. The 2-hop consistency increases by 0.150 on each dataset. The 1-hop gain is 0.100 on Cora and 0.050 on Flickr and HPPI. The gains across homophilic, heterophilic, and biological graphs provide broader evidence for the topological modulation. For the visualization, we use the Cora subset to show the sampled 1-hop and 2-hop connections.

Table 5. Connection Clustering Consistency on sampled subsets with and without topological modulation.

Dataset	1-hop			2-hop		
	Without	With	Δ	Without	With	Δ
Cora	0.250	**0.350**	+0.100	0.150	**0.300**	+0.150
Flickr	0.250	**0.300**	+0.050	0.200	**0.350**	+0.150
HPPI	0.250	**0.300**	+0.050	0.200	**0.350**	+0.150

In Fig. 4, the left panel shows the feature-based setting without topological modulation. The right panel presents the topology-aware setting. Both panels use the same sampled nodes and the same 1-hop and 2-hop connection pairs. Red solid lines denote 1-hop direct connections. Blue dashed lines denote 2-hop indirect connections. The topology-aware setting shows a clearer organization of the

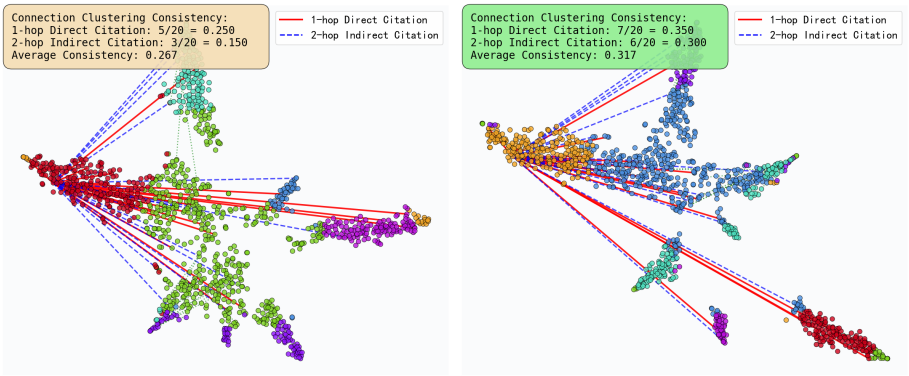


Fig. 4. Impact of topological modulation in tensor affinity on Cora. Red: 1-hop; Blue dashed: 2-hop.

predicted clusters. Several predicted groups appear more compact and easier to distinguish. More fixed connection pairs link nodes assigned to the same predicted cluster. The 1-hop consistency increases from 5/20 (0.250) to 7/20 (0.350). The 2-hop consistency increases from 3/20 (0.150) to 6/20 (0.300). The 1-hop support introduces immediate adjacency evidence into the distance terms. The 2-hop support adds indirect neighborhood evidence. The joint use of 1-hop and 2-hop support allows the tensor affinity to evaluate feature similarity together with local structural evidence. The combined structural evidence is intended to reduce the reliance of affinity estimation on feature proximity alone.

Embedding Similarity Structure Evaluation. We visualize cosine similarity heatmaps of the learned node embeddings to evaluate the class-related organization. A well-separated latent space is expected to exhibit a distinct block-diagonal pattern, with higher within-class similarity and lower cross-class similarity. All heatmaps share the same node ordering and similarity scale. Nodes are grouped by ground-truth labels. Red lines mark the class boundaries, and darker entries indicate higher pairwise similarity. The Ground Truth panel provides an ideal block-diagonal reference.

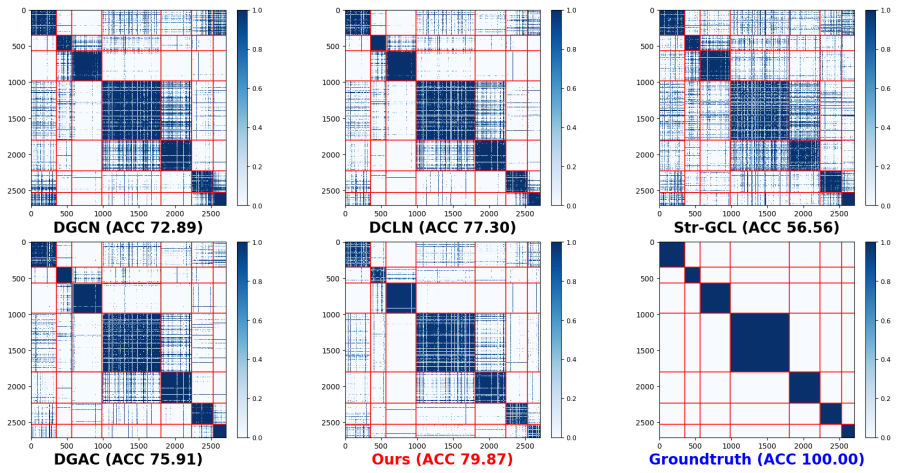
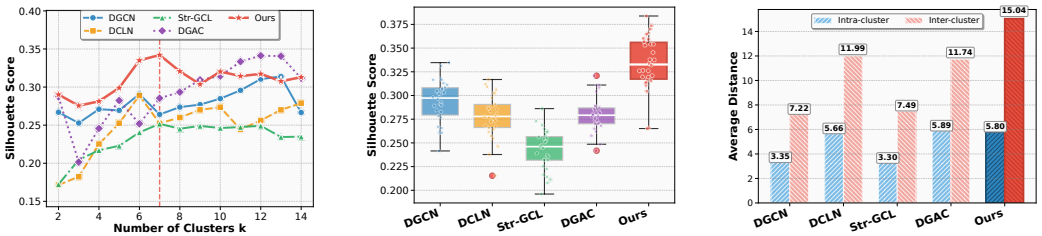


Fig. 5. Embedding similarity heatmaps on Cora. Red lines mark ground-truth class boundaries, and darker entries indicate higher pairwise similarity.



(a) Silhouette vs. k (b) Robustness under Perturbation (c) Intra/Inter-cluster Distances

Fig. 6. Cluster quality analysis on Cora: (a) Silhouette scores across different cluster counts, (b) Silhouette score distributions under bootstrap resampling, and (c) average intra- and inter-cluster distances.

Fig. 5 shows more pronounced off-diagonal similarities for DGCN and Str-GCL. Several class boundaries remain visually blurred. DGAC recovers the main diagonal blocks and exhibits relatively clear separation in several local diagonal regions. Noticeable cross-block similarity remains elsewhere. DCLN and TAGSL exhibit similar overall block contours and preserve the major class partitions. Compared with DGAC, TAGSL presents a more continuous central block and weaker off-diagonal similarities. TAGSL shows less scattered similarity than DCLN in several off-diagonal regions. TAGSL records the highest ACC of 79.87%, compared with 77.30% for DCLN and 75.91% for DGAC. The heatmaps qualitatively characterize the class-related organization, while ACC measures the overall agreement between the clustering and the ground-truth labels. The observed block organization is consistent with the exact spectral constraint and semantic alignment. The rank condition $\text{rank}(\boldsymbol{Q}^\top \boldsymbol{Q}) = n - c$ fixes the null-space dimension at c and aligns the spectral structure with the community count. Semantic alignment links the spectrally constrained structure to the embedding space. The exact spectral constraint and semantic alignment encourage the major class partitions to appear in the embedding similarities.

4.5 Separability and Clustering Evaluation

Although ACC and NMI rely on ground-truth labels, we further assess the intrinsic geometric separability of the learned embeddings via unsupervised criteria. This analysis examines how the embedding geometry reflects the target community structure. Ward's hierarchical clustering and K-means are first applied to the embeddings generated by TAGSL and the representative baselines DGCN, DCLN, Str-GCL, and DGAC. Specifically, Ward's clustering is used to calculate Silhouette scores across different cluster counts and under bootstrap resampling. K-means assignments are used to calculate average distances within and between clusters. The quantitative evaluation complements the preceding heatmap analysis by examining alignment with the target community count, stability under resampling, and geometric separation.

First, we perform hierarchical clustering over the cluster counts ($k \in [2, 14]$). In Fig. 6(a), the red dashed line marks the target community count on Cora ($c = 7$). TAGSL reaches its maximum Silhouette score at $k = 7$ and obtains the highest score among the compared methods at this cluster count. The compared baselines reach the maximum Silhouette scores at other cluster counts. This alignment is consistent with the structural partition encouraged by the exact spectral constraint.

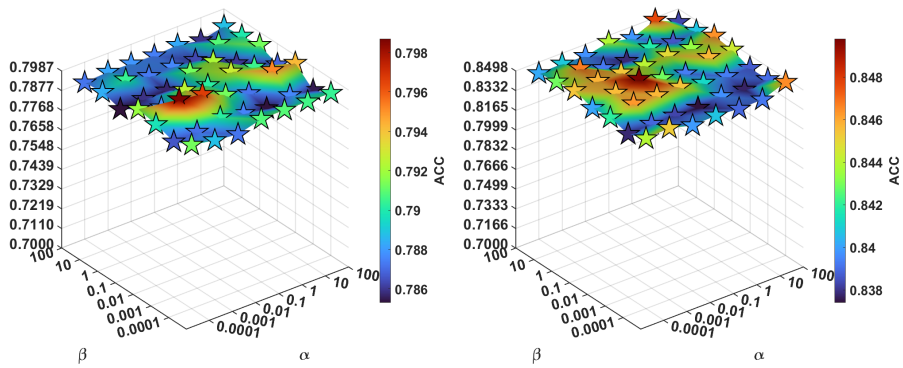
To assess the stability of embedding separability under bootstrap resampling, we conduct 30 iterations. For each iteration, a randomly sampled subset of the embeddings is clustered at $k = 7$. The distributions in Fig. 6(b) show that TAGSL achieves the highest median Silhouette score. Its lower quartile also lies above the medians of the compared baselines. DCLN and DGAC exhibit several low outliers, whereas the lower tail of TAGSL remains higher across the resampled subsets.

The elevated median and lower quartile indicate that TAGSL retains relatively high Silhouette scores across the resampled subsets, reflecting stable embedding separability.

Finally, we quantify compactness and separation in the latent space using average Euclidean distances within and between clusters based on K-means assignments. Fig. 6(c) shows that TAGSL records an average distance of 5.80 within clusters, close to DCLN (5.66) and DGAC (5.89). Its average distance between clusters reaches 15.04, the largest value among the compared methods. The result indicates that the geometric advantage of TAGSL is reflected in larger separation between clusters, while its compactness within clusters remains comparable to DCLN and DGAC. The above evaluation shows that TAGSL aligns its Silhouette peak with the target community count, retains relatively high scores under bootstrap resampling, and achieves separation between clusters.

4.6 Sensitivity to Structural Hyperparameters

This section investigates the sensitivity of **TAGSL** to the Laplacian smoothness weight α, the tensor affinity weight β, and the Nyström landmark size m. We first examine α and β on the homophilic Cora and heterophilic Flickr graphs. Both weights are varied over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. Silhouette Score is used for hyperparameter selection, while ACC is reported across the grid to visualize parameter sensitivity.



(a) Cora (b) Flickr

Fig. 7. Clustering accuracy under varying structural weights α and β on (a) Cora and (b) Flickr.

Fig. 7 shows that TAGSL maintains relatively high ACC across a wide range of α and β on both datasets. For Cora, TAGSL reaches 79.87% ACC in Fig. 7(a). The ACC surface shows smooth transitions across most of the parameter grid. Regions with strong performance extend along both the α and β directions, and neighboring settings produce similar results. The broad region exhibits limited ACC variation. The remaining local changes indicate that the balance between Laplacian smoothness and tensor affinity affects the clustering result. For Flickr, TAGSL reaches 84.98% ACC in Fig. 7(b). The surface contains more distinct rises and declines than the Cora surface, with clearer color transitions across neighboring settings. Regions with strong performance still appear under multiple combinations of α and β, although the distribution differs from that on Cora. The local ACC variations on Flickr differ from those on Cora. The comparison shows that TAGSL maintains relatively high ACC under multiple settings of α and β on both graphs, while the favorable regions differ between Cora and Flickr. Across the tested scales, ACC varies less on Cora than on Flickr. The local variations indicate that the balance between Laplacian smoothness and tensor affinity

affects clustering performance, especially on Flickr. The different variation patterns on Cora and Flickr support separate adjustment of α and β for each dataset.

Table 6. Effect of Nyström landmark size m on clustering accuracy, training time, and memory usage on HPPI. The selected setting is shown in bold.

Landmark Size (m)	ACC (%)	Time (s/epoch)	Memory (MB)
64	64.05	0.59	749
128	69.19	0.87	808
256	**71.59**	**1.45**	**975**
512	71.68	2.62	1395
1024	71.73	5.11	2188

We further investigate the impact of the landmark size m, which controls the size of the Nyström approximation and affects computational cost. Table 6 details the clustering accuracy and resource consumption on the HPPI network under varying m. As m increases from 64 to 256, ACC improves from 64.05% to 71.59%. The training time increases from 0.59 to 1.45 s/epoch, and memory usage increases from 749 to 975 MB. For $m \geq 256$, further increases in m yield marginal ACC improvement. ACC rises from 71.59% at $m = 256$ to 71.73% at $m = 1024$. Training time increases from 1.45 to 5.11 s/epoch, and memory usage grows from 975 to 2188 MB. Guided by the trade-off between performance and computational cost, we fix $m = 256$ across all primary evaluations. At $m = 256$, TAGSL achieves ACC close to the values obtained with larger landmark sizes while requiring less training time and memory.

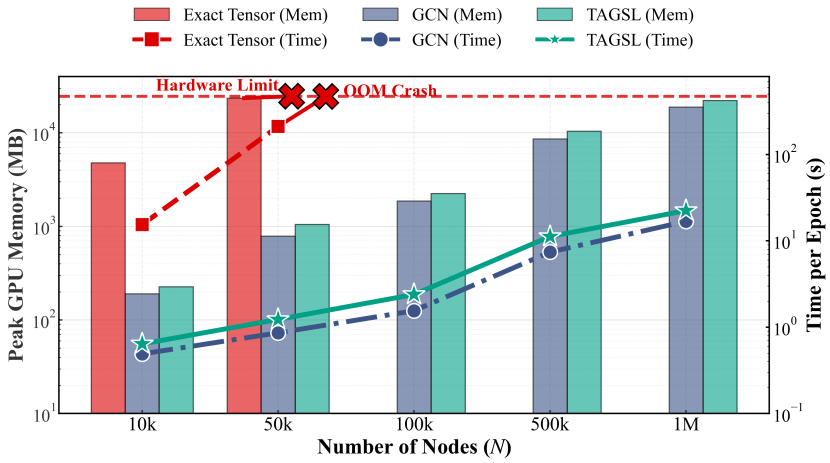
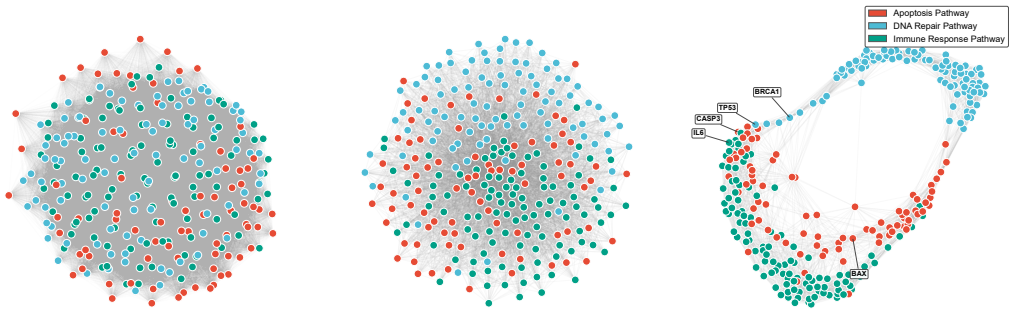


Fig. 8. **Scalability on synthetic graphs.** Exact Tensor rapidly hits the hardware limit at 100k nodes. **TAGSL** scales approximately linearly to 1M nodes with efficiency comparable to GCN, avoiding explicit dense tensor materialization.



(a) Raw Relational Database View (b) DGAC View (SS=0.039) (c) TAGSL View (SS=0.167)

Fig. 9. **Multi-Omics Database Cleaning on the STRING Network.** (a) Raw database topology corrupted by integration errors. (b) DGAC suffers from semantic confusion by propagating along noisy edges. (c) **TAGSL** adaptively refines graph structure under exact spectral constraints, materializing three relatively distinct semantic views while preserving biological crosstalk.

4.7 Efficiency and Scalability Analysis

To examine the empirical efficiency and scaling behavior of **TAGSL**, we evaluate the training time per epoch and peak GPU memory usage. We compare **TAGSL** with a lightweight GCN and an Exact Tensor implementation without low-rank approximation. The GCN serves as a lightweight message-passing reference, and Exact Tensor reflects the cost of explicit dense tensor computation.

As shown in Table 7, explicitly computing the exact high-order tensor affinity on the real-world HPPI network (40,887 nodes) encounters an Out-Of-Memory (OOM) failure. The result reflects the prohibitive memory cost of explicit dense tensor computation under the tested hardware configuration. In contrast, **TAGSL** uses 975 MB of GPU memory on HPPI. The training time of **TAGSL** incurs an additional 0.26 seconds per epoch compared with the lightweight GCN baseline. On Pubmed, **TAGSL** requires 0.73 seconds per epoch and 429 MB of GPU memory, whereas Exact Tensor requires 190.8 seconds per epoch and 15,097 MB. The additional computational cost is consistent with the $O(mNd)$ low-rank projection. The empirical results show that the proposed approximation avoids explicit dense tensor construction while maintaining manageable training time and memory usage on the evaluated real-world graphs.

Table 7. Training time (seconds per epoch) and memory consumption (MB). The exact tensor computation encounters OOM failures on HPPI.

Method	**Pubmed** (19,717 nodes)		**HPPI** (40,887 nodes)	
	Time	Mem	Time	Mem
Exact Tensor	190.8	15,097	**OOM**	**OOM**
GCN Baseline	0.61	258	1.19	704
TAGSL (Ours)	**0.73**	**429**	**1.45**	**975**

To investigate scalability at larger graph sizes, we construct a series of synthetic graphs with node sizes ranging from $N = 10k$ to $1M$. Guided by the hyperparameter analysis in Section 4.6, we fix the Nyström landmark size at $m = 256$. Fig. 8 reveals that the Exact Tensor approach rapidly

exhausts the hardware limit, triggering an OOM crash at 100k nodes. As N scales to 1 million, the time and memory trajectories of **TAGSL** exhibit approximately linear growth trends that closely parallel the GCN baseline.

The observed growth trend is consistent with the theoretical complexity analysis. The separable Nyström approximation reduces computation over the n^2-dimensional tensor representation. TAGSL further uses a quadratic surrogate for the embedding update. Fixing m restricts the core computational complexity to $O(mNd)$. The overall time complexity is $O\big(N(md + d^2) + |E|d\big)$. For fixed m and d on sparse graphs, the overall complexity grows linearly with the node count N. By integrating the Ky Fan reformulation with the separable Nyström strategy, **TAGSL** effectively circumvents explicit tensor instantiation and expensive eigen-decompositions. The framework successfully performs high-order tensor structure learning while limiting the asymptotic computational overhead to a manageable level, supporting linear scalability with respect to N.

4.8 Multi-Omics Graph Database Cleaning

In real-world multi-omics database management, integrating diverse sources such as experimental records and text mining introduces severe structural false positives. To evaluate our framework under extreme data integration noise, we extract a protein relational sub-network from the multi-omics STRING human database, targeting the apoptosis, DNA repair, and immune response pathways. We deliberately relax STRING's interaction confidence threshold to retain massive integration errors, and withhold all external node attributes to enforce a blind test scenario. Fig. 9(a) shows that the resulting relational view forms a highly entangled topology that substantially obscures the underlying biological functions.

We compare TAGSL with DGAC, a representative high-order baseline from Table 3. DGAC captures high-order semantics via graph diffusion across both topology and attribute graphs. The blind test setting removes reliable node attributes and reduces the effectiveness of the diffusion mechanism. The input topology contains false-positive integration edges, and graph diffusion can amplify structural noise by propagating semantics across erroneous inter-cluster links. Fig. 9(b) shows that DGAC retains substantial pathway mixing. In contrast, Fig. 9(c) shows that TAGSL reduces pathway mixing and materializes three relatively distinct semantic views. For quantitative comparison, we apply K-means with $k = 3$ to the learned embeddings and compute the SS via the resulting cluster assignments. TAGSL achieves an SS of 0.167, compared with 0.039 for DGAC. The apoptosis and immune response regions remain adjacent. Local overlap remains visible at the boundary between the apoptosis and immune response regions. The local overlap is consistent with biological crosstalk between the apoptosis and immune response pathways. Our framework optimizes a topology-aware tensor affinity alongside adaptive graph structure learning under exact spectral constraints. The affinity incorporates local structural densities to reduce the influence of spurious linkages during high-order modeling. The learned structural operator adapts to the evolving embedding geometry, and the rank condition supports exact graph structural decoupling.

Beyond clustering, TAGSL can also support automated semantic view materialization as an analytical capability for complex domain databases. We cross-reference the refined topology with curated biological annotations and observe qualitative consistency with domain expert knowledge despite the fully unsupervised setting. TP53 and BRCA1 are associated with the DNA repair view, IL6 with the immune response view, and BAX with the apoptosis view. The boundary positions of IL6 and CASP3 reflect interactions between the apoptosis and immune response pathways. The annotated positions of TP53, BRCA1, IL6, BAX, and CASP3 provide qualitative evidence that the refined topology retains biologically meaningful pathway organization and biological crosstalk.

5 Conclusion

This paper proposes **TAGSL**, a unified framework for DGC that couples high-order interaction modeling with adaptive graph structure learning under exact spectral constraints. The topology-aware tensor affinity reduces the influence of spurious linkages, while spectral rank constraints enable exact graph structural decoupling for semantic community formation. To address the computational bottlenecks of high-order modeling, TAGSL combines a separable Nyström approximation with Ky Fan reformulation. Under fixed landmark and embedding dimensions on sparse graphs, the overall complexity grows linearly with the node count. Extensive evaluations on diverse benchmarks demonstrate strong clustering performance against 13 representative baselines. Ablation and additional analyses further demonstrate structural coherence, embedding separability, and computational efficiency. Moreover, a real-world multi-omics case study shows reduced pathway mixing and relatively distinct semantic views qualitatively consistent with domain expert knowledge. Overall, the empirical evidence supports TAGSL as a scalable framework for knowledge discovery and topology refinement in graph databases. The current framework assumes a static observed graph and a known target community count, while the Nyström approximation uses a fixed landmark size. These settings focus TAGSL on static graphs with predefined community scale and a fixed balance between approximation quality and computational cost. Future work will extend TAGSL to evolving graphs and further adapt its community and landmark configurations during learning.

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A Derivation of the Tensor Affinity

We formulate the optimal affinity distribution using the Principle of Maximum Entropy [54]. Let the random variable x represent the internal feature divergence $\mathcal{E}_{\text{int}}$. We seek a probability density function $P(x)$ that maximizes the Shannon entropy:

$$\mathcal{H}(P) = -\int P(x) \ln P(x)\, dx, \tag{19}$$

subject to the normalization condition $\int P(x)\, dx = 1$. Following the hypothesis of local scaling [54], the expected internal divergence is governed by the local context scale $\mathcal{E}_{\text{ext}}$ imposing the constraint:

$$\mathbb{E}[x] = \int xP(x)\, dx = \mu\mathcal{E}_{\text{ext}}, \tag{20}$$

where $\mu > 0$ is a constant factor. We construct the Lagrangian functional $\mathcal{L}$ with multipliers γ_1 and γ_2:

$$\mathcal{L} = -\int P(x) \ln P(x)\, dx - (\gamma_1 - 1)\left(\int P(x)\, dx - 1\right) - \gamma_2\left(\int xP(x)\, dx - \mu\mathcal{E}_{\text{ext}}\right). \tag{21}$$

Taking the functional derivative with respect to $P(x)$ and setting it to zero yields $-\ln P(x) - \gamma_1 - \gamma_2 x = 0$. After normalization, this gives the exponential distribution $P(x) = \gamma_2 \exp(-\gamma_2 x)$ on $[0, \infty)$. Its expectation is $\mathbb{E}[x] = 1/\gamma_2$. To satisfy the local scaling constraint $\mathbb{E}[x] = \mu\mathcal{E}_{\text{ext}}$, we

obtain $\gamma_2 = 1/(\mu\mathcal{E}_{\text{ext}})$. The corresponding cumulative distribution is $F(x) = 1 - \exp[-x/(\mu\mathcal{E}_{\text{ext}})]$. By defining $\sigma = 1/\mu$ and evaluating it at $x = \mathcal{E}_{\text{int}}$, we obtain the tensor metric in Eq. (4).

B Design Rationale of Topological Modulation

The topology-aware modulation weights instantiate the internal divergence and local context on graph structures. The multiplicative $w^{(n)}_{ijkl}$ emphasizes joint consistency of the two target pairs and acts as a soft logical AND, suppressing the interaction when either pair lacks structural support. In contrast, the additive $w^{(d)}_{ijkl}$ aggregates cross-pair connectivity and acts as a soft logical OR, providing the local structural background used to normalize the target-pair divergence.

C Proof of the Coupling Theorem

Proof. We analyze the dependency structure by examining the mixed second-order partial derivative of the tensor metric $\mathcal{T} = 1 - \exp(-\sigma\mathcal{E}_{\text{int}}/\mathcal{E}_{\text{ext}})$. In a decoupled additive system the marginal rate of change with respect to one variable remains invariant to the other implying a vanishing mixed derivative. We compute the mixed derivative by first determining the gradient with respect to the internal divergence:

$$\frac{\partial \mathcal{T}}{\partial \mathcal{E}_{\text{int}}} = \frac{\sigma}{\mathcal{E}_{\text{ext}}} \exp\left(-\sigma \frac{\mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right). \tag{22}$$

Subsequently differentiating with respect to the context scale $\mathcal{E}_{\text{ext}}$ requires applying the product rule and chain rule:

$$\begin{aligned}
\frac{\partial^2 \mathcal{T}}{\partial \mathcal{E}_{\text{int}} \partial \mathcal{E}_{\text{ext}}} &= \frac{\partial}{\partial \mathcal{E}_{\text{ext}}} \left[\sigma \mathcal{E}_{\text{ext}}^{-1} \exp\left(-\sigma \mathcal{E}_{\text{int}} \mathcal{E}_{\text{ext}}^{-1}\right)\right] \\
&= -\sigma \mathcal{E}_{\text{ext}}^{-2} \exp\left(-\sigma \frac{\mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right) + \frac{\sigma}{\mathcal{E}_{\text{ext}}} \exp\left(-\sigma \frac{\mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right) \left(\sigma \mathcal{E}_{\text{int}} \mathcal{E}_{\text{ext}}^{-2}\right) \\
&= -\frac{\sigma}{\mathcal{E}_{\text{ext}}^2} \left(1 - \frac{\sigma \mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right) \exp\left(-\sigma \frac{\mathcal{E}_{\text{int}}}{\mathcal{E}_{\text{ext}}}\right).
\end{aligned} \tag{23}$$

For finite energy states and $\sigma > 0$, this derivative is generally non-zero. The resulting cross-term shows that local structural context modulates the sensitivity of the tensor metric to internal divergence; accordingly, local topology controls target-pair interaction strength through the contextual scale, establishing the stated non-separable coupling. □

D Convergence Analysis of Structural Alternating Optimization

The complete TAGSL objective is $\mathcal{J} = \mathcal{L}(\boldsymbol{H}, \boldsymbol{Q}) + \mathcal{L}_{\text{cluster}}$. The network parameters $\boldsymbol{W}$ and cluster centroids $\boldsymbol{\mu}$ are updated through standard backpropagation on the joint objective $\mathcal{J}$. The convergence analysis focuses on the constrained structural sequence $\{(\boldsymbol{H}^{(t)}, \boldsymbol{Q}^{(t)})\}_{t\geq 0}$ generated by the alternating updates of $\mathcal{L}(\boldsymbol{H}, \boldsymbol{Q})$.

Regularity Conditions. We assume that (i) $(\boldsymbol{H}^{(t)})^\top \boldsymbol{H}^{(t)} = \boldsymbol{I}$ for all iterates, so $\{\boldsymbol{H}^{(t)}\}$ remains on the compact Stiefel manifold; (ii) scale normalization keeps $\{\boldsymbol{Q}^{(t)}\}$ bounded; (iii) the block gradients with respect to $\boldsymbol{Q}$ and $\boldsymbol{H}$ are Lipschitz continuous on the bounded feasible domain; and (iv) η_Q and η_H satisfy the corresponding blockwise descent conditions. The auxiliary matrix $\boldsymbol{F}^{(t)}$ and Nyström basis $\boldsymbol{P}_n$ are treated as fixed during their respective block updates. These conditions imply that the continuous structural objective $\mathcal{L}(\boldsymbol{H}, \boldsymbol{Q})$ is lower bounded on the feasible domain.

Lemma 2 (Sufficient decrease of the Q-update). *For fixed $H^{(t)}$, $F^{(t)}$, and $\lambda^{(t)}$, define the local structural subproblem*

$$\phi_t(Q) = \alpha\|QH^{(t)}\|_F^2 + \lambda^{(t)}\|QF^{(t)}\|_F^2 + \frac{\rho}{2}\|QF^{(t)}\|_F^4. \tag{24}$$

Let $\mathcal{Q} = \{Q : Q\mathbf{1} = \mathbf{0}\}$. For a sufficiently small learning rate $\eta_Q > 0$, the projected update

$$Q^{(t+1)} = \Pi_{\mathcal{Q}}\left(Q^{(t)} - \eta_Q \nabla\phi_t(Q^{(t)})\right) \quad \textit{satisfies} \quad \phi_t(Q^{(t+1)}) \le \phi_t(Q^{(t)}) - c_Q\|Q^{(t+1)} - Q^{(t)}\|_F^2 \tag{25}$$

for a constant $c_Q > 0$.

Proof. The gradient of ϕ_t is Lipschitz continuous on the bounded domain of Q. The standard descent lemma therefore guarantees sufficient decrease for the unconstrained gradient step when η_Q is smaller than the reciprocal of the corresponding Lipschitz constant. The row-wise mean-centering operation is the Euclidean projection $\Pi_{\mathcal{Q}}(Z) = Z(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)$, which satisfies $\Pi_{\mathcal{Q}}(Z)\mathbf{1} = \mathbf{0}$. The non-expansiveness of the projection preserves the sufficient-decrease property and yields Eq. (25). □

The dual update $\lambda^{(t+1)} = \lambda^{(t)} + \rho\|Q^{(t+1)}F^{(t)}\|_F^2$ penalizes spectral null-space violations, while Lemma 2 concerns the primal Q-update for fixed $F^{(t)}$ and $\lambda^{(t)}$.

Lemma 3 (Feasibility and descent of the H-update). *For fixed $Q^{(t+1)}$, let the differentiable structural objective used by the embedding update be $\mathcal{L}(H, Q^{(t+1)})$. The objective includes the Nyström-based structural approximation described in the main paper. For a sufficiently small learning rate $\eta_H > 0$, there exists a constant $c_H > 0$ such that the gradient step followed by QR retraction satisfies*

$$(H^{(t+1)})^\top H^{(t+1)} = I, \mathcal{L}(H^{(t+1)}, Q^{(t+1)}) \le \mathcal{L}(H^{(t)}, Q^{(t+1)}) - c_H\|H^{(t+1)} - H^{(t)}\|_F^2. \tag{26}$$

Proof. The embedding constraint defines the compact Stiefel manifold $\mathcal{M}$. The gradient of $\mathcal{L}(H, Q^{(t+1)})$ is Lipschitz continuous on $\mathcal{M}$. The QR decomposition of the tentative update produces an orthonormal factor, which gives the feasibility relation in Eq. (26). Under the stated learning-rate condition, the descent property of a QR-retracted gradient update follows from the Riemannian descent lemma [1], yielding the second relation in Eq. (26). □

Theorem 4 (Convergence of the structural alternating sequence). *Suppose that the regularity conditions above hold and the auxiliary updates preserve the blockwise sufficient-decrease and relative-error conditions of the alternating procedure. Then $\{\mathcal{L}(H^{(t)}, Q^{(t)})\}_{t\ge 0}$ converges, every accumulation point of $\{(H^{(t)}, Q^{(t)})\}_{t\ge 0}$ is a critical point of $\mathcal{L}(H, Q)$, and the entire structural sequence converges to a critical point when $\mathcal{L}(H, Q)$ satisfies the Kurdyka–Łojasiewicz property.*

Proof. Lemmas 2 and 3, together with lower boundedness of the structural objective, imply convergence of the objective values and

$$\sum_{t=0}^{\infty}\left(\|Q^{(t+1)} - Q^{(t)}\|_F^2 + \|H^{(t+1)} - H^{(t)}\|_F^2\right) < \infty. \tag{27}$$

Thus successive structural updates vanish, and the blockwise optimality and relative-error conditions imply that every accumulation point is critical. Since $\mathcal{L}(H, Q)$ consists of algebraic operations, squared norms, and exponential terms, it belongs to the log-exp definable class and satisfies the Kurdyka–Łojasiewicz property [4]. The standard KL argument then yields convergence of the entire structural sequence to a critical point. □

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